

D-5 lift ablation -- CJaC-grounded vs. raw frontier models: pre-registration

Phase EXPERIMENT item 7 of the 2026-09-05 LOCK directive; first D-5 data point per DIRECTION_D_ROADMAP.md. DESIGN ONLY -- no execution until the item set is frozen and Andy approves the budget. Copyright 2026 Andrew M Cohen. Apache 2.0.

Status: PRE-REGISTERED 2026-09-05 (design); **PROMOTED to demo-gating and resequenced ahead of the configuration ablation by the 2026-09-05 Addendum (s.3)**; amended 2026-09-14 (budget itemized per arm; grounded-system designation; replay-first accounting). **APPROVED 2026-09-14 -- Andy: "i approve the \$30 lift."**
ITEM SET FROZEN 2026-09-14: scripts/experiments/lift_items_v1.json, 24 items, sha256 d4c34178b8f74998da259bd5868858d11e71dfd2dcaa3c95b9b0c07bf4fb833, recorded in scripts/ci/frozen_artifact_manifest.json (CI-enforced) BEFORE any arm ran; the runner (scripts/experiments/run_lift_ablation.py) refuses --live if the hash drifts. Review copy for Andy: **double-click** review/D5_LIFT_ITEMS_V1.pdf (one item per page, notes field). Execution status: **ALL SIX ARMS RUN 2026-09-15** (section 11); results provisional pending Andy's audit.

1. Question

On consumer-debt questions inside the v1.0 corpus's scope, how much better does a model answer WITH CJaC's rules attached than the same model answers on its own -- and does the grounded system abstain correctly (ask for the missing dispositive fact) where the raw model guesses?

2. Arms (6)

Arm	System	Rules attached
G-A	claude-opus-5	the relevant v1.0 node(s) as context, with the completeness checklist and the instruction to abstain and ask when a dispositive fact is missing
G-O	gpt-5.5	same
G-G	gemini-2.5-pro	same
R-A	claude-opus-5	none -- the question alone, same abstention instruction
R-O	gpt-5.5	none
R-G	gemini-2.5-pro	none

Same prompt shell, same temperature (0), same item text, model identifiers frozen at execution. Node selection for the G arms is by the item's declared depends_on_node_ids (as in scenarios.json), not by retrieval -- retrieval quality is a separate question and is held constant by construction.

3. Items

• The 5 existing demo scenarios (scripts/corroboation/scenarios.json), each expanded into a concrete fact pattern.

• A Cowork-drafted expanded set of **15-20 items** across the 19 nodes (AS FROZEN 2026-09-14: 19 new items + the 5 expanded scenarios = 24 items covering all 19 nodes; mix 8 answerable / 5 abstain-correct / 11 trap -- traps over-weighted because predictions L1-L3 locate the value there; the answerable items are the control), three item types in fixed proportion: **(i) answerable** -- all dispositive facts given, one correct answer derived from v1.0 content; **(ii) abstain-correct** -- a dispositive fact deliberately withheld, the correct response is to identify and ask for it; **(iii) trap** -- a fact pattern where the folk-legal answer is wrong and the v1.0 node says why (drawn from the dangerous-direction ledger rows: e.g., "they never sent a signed contract so it's the 2-year period", "the

judgment is 10 years old so it's dead", "single vehicle so nothing to file").

• Each item carries: facts, the question, the v1.0 node(s) it depends on, the ground-truth answer or the ground-truth missing fact, and the DD flag. Ground truth is derived ONLY from frozen v1.0 content; where the node's answer depends on a GLOSS-FOR-COUNSEL proposition, the item is excluded until the counsel ruling.

• **Freeze:** the item file is committed with its sha256 recorded here and in `frozen_artifact_manifest.json` BEFORE any arm runs. Items are burned after one use for headline reporting (house rule); re-use is allowed only for D-5 trend tracking against new model generations, stated as such.

4. Scoring (abstention-credit, house rules)

Response	Answerable item	Abstain-correct item
Correct answer (matches ground truth)	1	--
Correct abstention (names the missing dispositive fact)	0.5	1
Abstains without naming the fact / generic "see a lawyer"	0.25	0.5
Wrong answer, safe direction (overstates the consumer's position)	0	0
Wrong answer, DANGEROUS direction	0, and counted separately	0, counted separately

Scored by an LLM judge from a family not under test for that item (rotating), with rubric and ground truth in the judge prompt; **Andy audits 20% of judgments (all DD-wrong calls plus a random sample)**; results dual-reported (raw judge, audited). The DD-wrong COUNT is reported alongside the score -- a system that scores 0.7 with zero dangerous-direction errors and one that scores 0.7 with six are not the same system.

5. Lift

$\text{Lift}(\text{model}) = \text{mean score}(\text{G-model}) - \text{mean score}(\text{R-model})$, per model and pooled, with 95% bootstrap intervals over items; reported per item type. The headline "lift number" is the pooled figure with its basis stated in the same sentence (n items, item mix, judge, audit rate, model versions, date). Also reported: DD-wrong counts per arm; abstention precision (of abstentions, share that named the right fact); the per-item table.

6. Predictions (pre-registered)

L1. Lift > 0 for all three models, largest on trap items. L2. Raw models abstain less and produce more DD-wrong answers on abstain-correct items. L3. Lift on answerable items is smaller for the strongest raw model; the corpus's value concentrates in abstention and traps.

7. Budget and gating (itemized per arm -- Addendum s.1 replay-first rule)

Nothing in this experiment can run on recorded fixtures: every arm requires fresh model output on items that did not exist before. So every call below is live, itemized, and gated.

Arm	Calls	Est. tokens/call (in+out)	Est. \$
G-A grounded claude-opus-5	25	~6K in (node context) + 1K out	~\$3.50
G-O grounded gpt-5.5	25	same	~\$3.00
G-G grounded gemini-2.5-pro	25	same	~\$1.50
R-A raw claude-opus-5	25	~0.5K in + 1K out	~\$2.00
R-O raw gpt-5.5	25	same	~\$1.50
R-G raw gemini-2.5-pro	25	same	~\$0.75

Arm	Calls	Est. tokens/call (in+out)	Est. \$
Judge (rotating family, 150 judgments)	150	~3K in + 0.3K out	~\$6.00
Smoke (3 items x 6 arms + judge)	24	--	~\$2.00

Total estimate ~\$20; proposed cap \$30 (APPROVED 2026-09-14). Runner's own pre-run estimate on the frozen 24-item set (price table of record printed with every run; chars/4 input, 900-token answers, 350-token judgments): grounded batch \$11.33 (G-A 5.63, G-O 3.12, G-G 2.58), raw batch \$6.25 (R-A 2.86, R-O 1.28, R-G 2.12), smoke \$2.03; total ~\$19.60 (single tranche; per-run cap \$15 respected by running arms as two batches of three). Runs are Andy's, smoke first. Preconditions: item file frozen with hash; calibration green; measurement-of-record recorded (done 2026-09-05). **"The grounded system" for the headline** is arm G-A (claude-opus-5 with the v1.0 node attached) -- the configuration the demo skill runs; G-O and G-G are reported alongside as robustness checks, not as the headline.

8. Known limitations (stated wherever results are reported)

One corpus, two states plus the federal spine; items authored by the same team that authored the corpus (mitigated by the trap design and the audit, not eliminated); LLM-judged with partial attorney audit; node selection by declaration rather than retrieval; model versions of record only. Findings about v1.0 content surfaced by this experiment go to POST_V1_BACKLOG.md.

9. Deliverable

docs/experiments/results/D5_LIFT_V1.md: the lift table, the DD-wrong counts, the per-item results, the audit reconciliation, and the one-paragraph basis statement for the messaging.

10. Execution protocol (added 2026-09-14 on approval)

Runner: scripts/experiments/run_lift_ablation.py (reuses the corroboration runner's key loading, JSON parsing and model identifiers; temperature 0 where the API accepts it; judge rotation A->O, O->G, G->A; scoring table of section 4 encoded; bootstrap CIs over items; dual-report writer; audit-sample export). Dry-run exercised end to end 2026-09-14 (144 synthetic judgments; no API calls). No corroboration-runner change in this round (one-variable rule holds).

Order, all runs Andy's, from the repo root, each under the \$15 per-run cap:

1. `python3 scripts/experiments/run_lift_ablation.py --live --smoke -- 3 items x 6 arms + 18 judgments, ~$2.` Confirms keys, JSON parsing on all three families, judge parsing. Inspect the printed categories; if any arm returns UNSCORED for all three items, stop and report before spending more.
2. `python3 scripts/experiments/run_lift_ablation.py --live --batch grounded -- G-A, G-O, G-G, 24 items, ~$11.`
3. `python3 scripts/experiments/run_lift_ablation.py --live --batch raw -- R-A, R-O, R-G, 24 items, ~$6.`
4. `python3 scripts/experiments/run_lift_ablation.py --aggregate --live-only --` writes docs/experiments/results/D5_LIFT_V1.md (raw-judge and audited columns) and docs/experiments/results/lift_v1/AUDIT_SAMPLE.md (every DD-wrong call + a seeded random 20%). Commit the run JSONs, the summary and the audit sample (GitHub Desktop).
5. Andy's audit: rule on each audit-sample entry (confirm / override); Cowork enters the rulings as audited_category / audit_note in the run JSON and re-runs step 4. The audited column is the column of record; the raw-judge column is always reported beside it.

The smoke run's records are excluded from aggregation by default (`--include-smoke` to override) so the headline is computed on the full batches only. Smoke + batches use the same frozen items, so the three smoke items are

answered twice by each arm; the batch answer is the one of record.

Budget accounting: envelope \$250; ~\$165 spent through the measurement of record; this experiment ~\$20 (cap \$30) -> ~\$185-195 after completion. The configuration ablation remains unfunded pending the trim decision.

11. Execution record and provisional results (2026-09-15; pre-audit)

Runs (all Andy's): smoke run_20260915T005504Z (3 items x 6 arms, \$3.54 est.); grounded batch run_20260915T140855Z (G-A 24, G-O 24, G-G 17 -- halted at the \$15 per-run cap, \$15.11); G-G completion run_20260915T152051Z (7 items, \$0.59); R-A run_20260915T152510Z (~\$8); R-O + R-G run_20260915T154855Z (\$3.46). **Estimated spend \$26.79 + smoke \$3.54 = ~\$30.3** (runner's price table; the design's ~\$20 estimate assumed 900-token answers -- the models wrote 2-3x that, and claude-opus-5's thinking tokens count as output). Andy approved the overage ("budget approved") before the raw arms ran. Envelope after this experiment: ~\$195 of \$250.

Errata found after the freeze (item file untouched; hash unchanged):

- **L01 (item):** the facts put service on 2026-09-01 (answer due 09-15) but the reference date is 2026-09-17, so the deadline had already passed; my ground truth treated it as future and the rubric listed "deadline has passed" as the dangerous error. Every arm was judged wrong_dangerous for a correct statement. Excluded from the raw-judge headline (n = 23); scored in the audited column on Andy's ruling against the corrected ground truth (results doc, Errata).
- **J-1 (judge prompt):** the judge was not told the reference date, so gpt-5.5 read the L24 answers' correct "today is 2026-09-17" arithmetic as an error (G-A/L24, R-A/L24 both marked wrong_dangerous). Fixed in the judge prompt for future runs; v1's raw-judge column stands as produced and the audit resolves it.
- **J-2 (judge legal error):** claude-opus-5 judging G-G/L16 asserted a 10-day CCP 703.520 deadline; the node's quoted text says 15/20 days, which is what the answer said.
- **J-3 (classification):** gemini judging G-O/L17 classed an over-cautious abstention as wrong_dangerous; Andy's call.
- **G-A/L16:** the model's answer JSON had an unescaped quote and did not parse, so no judgment was made; the answer is intact in answer_raw and is recovered by --rejudge (~\$0.10) with a lenient parser now used on the live path too.

Provisional raw-judge results (n = 23, L01 excluded, before the audit and before the G-A/L16 re-judge):

Arm	mean score (95% CI)	DD-wrong	answerable / abstain-correct / trap
G-A (headline grounded system)	0.909 [0.773, 1.0] (22 scored)	1	0.875 / 1.0 / 0.889
G-O	0.913 [0.783, 1.0]	2	0.875 / 1.0 / 0.9
G-G	0.913 [0.783, 1.0]	0	1.0 / 0.8 / 0.9
R-A	0.783 [0.609, 0.957]	3	0.75 / 1.0 / 0.7
R-O	0.913 [0.783, 1.0]	1	1.0 / 0.8 / 0.9
R-G	0.913 [0.783, 1.0]	2	0.875 / 1.0 / 0.9

Paired lift (G minus R): claude-opus-5 **+0.136 [-0.045, +0.318]**; gpt-5.5 0.0 [-0.174, +0.174]; gemini-2.5-pro 0.0 [-0.174, +0.174]; pooled **+0.044 [-0.059, +0.147]**. DD-wrong: grounded 3 vs raw 6.

Reading (provisional; the audited column is the column of record once Andy has ruled):

- **L1 (lift > 0 for all three, largest on traps): NOT SUPPORTED.** Only the claude pair shows a positive point estimate, and its interval includes zero. gpt-5.5 and gemini-2.5-pro raw score the same as grounded on this set.
- **L2 (raw models abstain less / more DD-wrong on abstain-correct items): NOT SUPPORTED on abstention** -- every arm handled 4-5 of the 5 abstain-correct items; **weakly supported on DD-wrong overall** (6 vs 3, small counts, several of them judge-disputed).

• **L3 (value concentrates in abstention and traps): PARTLY.** Where the corpus demonstrably mattered, it was precise-rule items, not folk-legal traps: **L15** (CA single-vehicle "automatic" exemption -- all three raw arms gave a wrong 10-day deadline or denied the automatic proceeds rule; all three grounded arms right), **L03** (Reg F five-business-day mailbox assumption -- two raw arms early; all grounded right), and **L01** on the limitations half (two raw arms computed from last/first-missed payment and called the suit barred; all grounded arms computed from charge-off). The folk-legal traps (Henson, "never signed = 2-year", brokerage under the \$50k cap, county-court-vs-JP) did not trap current frontier models at all.

• **Honest headline for the messaging, pending audit:** "On 23 frozen consumer-debt items, attaching CJaC's v1.0 rules to claude-opus-5 raised its score from 0.78 to 0.91 and halved dangerous-direction errors across models (6 to 3), but the pooled lift is +0.04 with a confidence interval that includes zero; the demonstrable value is on precise-rule items (deadline computation, statutory dollar/day figures) where raw models were confidently wrong." That is the claim the data supports; nothing stronger.

• **Lesson for a v2 item set (POST_V1_BACKLOG):** traps must be calibrated against what current frontier models actually get wrong (deadline arithmetic, statutory figures, rule-specific mechanics), not against folk-legal errors; every item must be date-checked against the reference date mechanically; the judge prompt must carry the reference date (done).

Next: Andy runs --rejudge (G-A/L16) and --aggregate --live-only, commits, then rules on the audit sample (review/D5_LIFT_AUDIT_SAMPLE.pdf: 41 entries = all DD-wrong + errata + judge-flagged + a seeded random 20%). Cowork enters the rulings as audited_category, re-aggregates, and the audited column becomes the column of record.

D-5 rehearsal kit -- three scenarios with the grounded system's expected answer

REHEARSAL ONLY -- NOT EVIDENCE -- NO EXTERNAL CLAIMS. These three prompts exist so Andy can paste them into raw frontier models himself and see the contrast before the pre-registered lift number exists. Nothing observed this way is recorded, cited, or quoted anywhere. The expected answers are derived from the frozen debt-demo-v1.0 nodes named on each card; they are what the grounded system SHOULD say, not a claim that any model does say it. (Addendum 2026-09-05, s.3a. Copyright 2026 Andrew M Cohen. Apache 2.0.)

How to use. Paste the block under "Prompt" into a raw model with no other context. Compare to the "Expected grounded answer." Watch for three things: (1) does the raw model get the arithmetic and the rule right; (2) does it ASK for the missing fact where the grounded answer asks; (3) does it say anything in the dangerous direction (safe / no claim / out of time when the opposite is true).

Scenario 1 -- California COVID tolling on a credit-card statute of limitations

Prompt

I live in California. I stopped paying a Chase credit card and the account went into default on January 15, 2018. A debt buyer sued me on it on May 1, 2022. Is the lawsuit too late under the statute of limitations? What should I do about it?

Expected grounded answer (nodes: CA-SOL-WRITTEN-CONTRACT-DEBT)

• The period is four years for a written contract or a credit-card book account / account stated (CCP 337(a), 337(b)). Untolled deadline: January 15, 2022.

• That deadline had NOT passed as of April 6, 2020, so Judicial Council Emergency Rule 9(a) tolled it for 178 days (April 6 - October 1, 2020): new deadline about July 12, 2022. A suit filed May 1, 2022 is **timely** -- the raw "it's been more than four years" answer is wrong in the dangerous direction for the creditor's timing but the CONSUMER-facing danger runs the other way: a consumer told "too late, ignore it" would default.

• Compare the FILING date to the deadline, not today.

• Even where a suit is late, limitations is an affirmative defense that must be PLEADED in the answer (CCP 458; 431.30(b)(2)) -- a timely bare denial waives it.

• Grounded system should ask: any payment or signed acknowledgment after default (CCP 360)? Any period out of state (CCP 351, with the constitutional caveat)? A choice-of-law clause (CCP 361 / *Resurgence*)? Has the debtor died (CCP 366.2 -- one year from death)?

Citations the grounded answer carries: Cal. Code Civ. Proc. 337(a)-(b), 350, 360, 361, 458; Cal. R. Ct., emergency rule 9(a) (as amended May 29, 2020).

Scenario 2 -- Texas bank-account freeze: what is actually protected

Prompt

I'm in Texas. A debt buyer got a judgment against me on an old credit card and my bank just froze my checking account. It has \$2,900 in it: \$1,800 is my Social Security that's direct-deposited every month, and \$1,100 is the paycheck my employer direct-deposited last Friday. I thought Texas doesn't allow garnishment. What can they take?

Expected grounded answer (nodes: TX-WAGE-GARNISHMENT-PROHIBITION, TX-EXEMPT-PERSONAL-PROPERTY)

• Texas bars garnishment of CURRENT WAGES for ordinary debts (Tex. Const. art. XVI, sec. 28; CPRC 63.004) -- but that protects wages in the employer's hands. Once the paycheck is deposited, Texas has no general bank-account exemption, and whether deposited wages keep their character is unsettled (counsel gloss). The \$1,100 is at risk.

• The \$1,800 of directly deposited Social Security is protected: 42 U.S.C. 407(a), and under 31 C.F.R. 212 the bank must itself protect two months of directly deposited federal benefits before honoring the garnishment (the "lookback"). The consumer should still assert it.

• Texas exemptions are NOT self-executing: file an answer / motion to dissolve in the garnishment proceeding (TRCP 664a) asserting the exemptions -- do not wait.

• Grounded system should ask: is the account joint with a spouse (Fam. Code 3.202 marital-property screen)? Any unemployment or workers' compensation deposits (Lab. Code 207.075, 408.201)? Was the underlying judgment validly served (default set-aside routes)?

• Dangerous-direction trap the raw model may fall into: "Texas doesn't allow garnishment, so the freeze is illegal" (wrong -- the wage bar does not reach deposited funds) OR "the whole account is fair game" (wrong -- the Social Security is protected).

Citations: Tex. Const. art. XVI, sec. 28; Tex. Civ. Prac. & Rem. Code 63.004, 31.0025; 42 U.S.C. 407(a); 31 C.F.R. 212.6, 212.3; Tex. Prop. Code 42.001-42.002; Tex. Lab. Code 207.075.

Scenario 3 -- FDCPA validation rights and a collector that keeps calling

Prompt

A collection agency sent me a letter about a medical bill and I got it on September 1. I don't think I owe it, so on September 20 I mailed them a letter saying I dispute the debt and asking them to prove it. They never sent me anything back, but on September 25 they called me and told me to pay or they'd send it to a lawyer. They've called four more times since. What are my rights?

Expected grounded answer (nodes: FDCPA-VALIDATION-NOTICE-1692g; FDCPA-REGF-CALL-FREQUENCY-1006.14b; FDCPA-COVERAGE-DEBT-COLLECTOR-1692a6)

• Coverage first: a third-party agency collecting a consumer medical debt is a "debt collector" (15 U.S.C. 1692a(6)) and the bill is a "debt" (1692a(5)).

- • The consumer's WRITTEN dispute, sent within the 30-day validation period (received September 1 + 30 days; mailed September 20), triggers 1692g(b): the collector "shall cease collection of the debt ... until" it mails verification. The September 25 call and the four after it are collection activity during the suspension -- each is a 1692g(b) violation regardless of how many calls were made (the 7-in-7 count is beside the point).
 - • "Pay or we'll send it to a lawyer" while the debt is disputed and unverified may also be a 1692e(5)/(10) misrepresentation if no such action was intended, and calls that continue after a dispute must note the debt is disputed in any credit reporting (1692e(8)).
 - • Remedies: actual damages, statutory damages up to \$1,000, fees (1692k(a)); one year from each violation (1692k(d), *Rotkiske*). Keep the call log; keep proof of mailing.
 - • Grounded system should ask: exact date the dispute was RECEIVED (sent-vs-received is a counsel item); whether the letter was the first communication and whether it contained the 1692g(a)/Reg F 1006.34 notice; any bankruptcy; whether the consumer's state adds a longer period.
 - • Dangerous-direction trap: "they're allowed to call up to seven times a week" (true in general, irrelevant here -- the dispute suspends ALL collection).
- Citations:** 15 U.S.C. 1692a(5)-(6), 1692e(5), (8), (10), 1692g(a)-(b), 1692k(a), (d); 12 C.F.R. 1006.34, 1006.38; *Rotkiske v. Klemm*, 589 U.S. 8 (2019).
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Again: REHEARSAL ONLY. The formal comparison is D5_LIFT_ABLATION_DESIGN.md, pre-registered, with frozen items, a rotating judge, an attorney audit, and dual-reported results. Nothing seen with these three prompts is a result.